Izmir Institute of Technology
Electrical and Electronics Eng. Dep.

# EE 471 Modern Software Development Practices and Technologies

# In-class exercise Week #9

Thursday 30/04/2026

**Preliminary Task:** Study the shared course materials Mediapipe solutions

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**Task 1:** Choose 3 of the following vision tasks: Object detection, Image classification, Image segmentation, Hand landmark detection, Face landmark detection, and Pose landmark detection

Read through their documentation, and demonstrate them in Python.

**Deliverables (Deadline: end of the lecture):**

1) The link for your public github repository
2) A screen recording video demonstrating each of the tasks
3) (Bonus) Complete all 6 of the tasks.

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**Task 2:** Use a pose estimation library (e.g. mediapipe) to detect body pose in a given image, and write a script to classify which arm is up. Using Docker/cog containerize it.

**Input:** $PATH-TO-INPUT-IMAGE, **Output: "**left", "right", "both" or "None"
Test your script with 3 input images provided: **pose-1.jpg, pose-2.jpg, and pose-3.jpg.**

**Task 3:** Use a face detection library (e.g. mediapipe) to detect the face's looking direction in a given image, and write a script to classify if the face is looking towards left, right, or straight. Using Docker/cog containerize it.

**Input:** $PATH-TO-INPUT-IMAGE, **Output: "**left", "right", or "straight"
Test your script with 3 input images provided: **face-1.png, face-2.png, and face-3.png.**

**Deliverables (Deadline: until midnight):**

1) The link for your public github repository
2) A screen recording video demonstrating each of the tasks
3) A screen recording video with you explaining the purpose of each of the code blocks used (line by line **not needed**, but still try to explain thoroughly).